

EVAN LANE

WIRELESS HACKING



HOW TO HACK

WIRELESS NETWORKS
BEGINNER GUIDE

WIRELESS HACKING 101

Wireless Hacking

How to Hack Wireless Networks Beginner's Guide

Evan Lane

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Printed in the USA

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Everyone Suffers From

Introduction

Hacking is something that everyone has a general curiosity about. People want to know what it is that attracts so many people to wanting to do hacking whether it is legally or illegally.

Hacking is one of those things that is feared but holds people's attention because of the myths and various rumors that are surrounding the topic.

It does not matter what it is that you are wanting to use hacking for, with this book, you are going to learn how you can hack into a wireless network as a beginner. All of the steps in this novel

are set into place to assist you in something that you are interested in, in the best way possible.

There are plenty of books on this subject on the market, thanks again for choosing this one! Every effort was made to ensure it is full of as much useful information as possible, please enjoy!

Please note that all of the content that is in this book is for educational purposes only and is not meant to be used in any way that is considered to be illegal.

Hacking is highly illegal and not only punishable with fines, but with time in prison as well. Please do not hack into anything without the expressed permission of the system's owner and

make sure that you get the permission in writing so that you can have some protection in case the owner decides to try and get you in trouble for it. Should you have trouble getting the permission of the system's administrator, then you can always set up a virtual environment and hack your own system!

Chapter 1: Before You Hack

Hacking is not a skill that you are born with. It is a skill that has to be learned over an extended period of time so that you can actually become good at it. It doesn't matter if you are hacking as a white hat hacker or as a black hat hacker you are going to have to go through a process that is involved in making sure that you have the skills it requires in order to actually do what you are wanting to do. (Please remember that any hacking that is not done with the network's administrator's permission is highly illegal. **Do not do it!**)

The biggest thing that you're going to

want to remember when you're going through and learning the skills that it takes to be a hacker is that you're going to need to be patient. It is not going to be something that you learn overnight.

Patience is a must for when you are hacking because if you are not patient, you may end up doing something that you do not necessarily want to do. That or you will end up missing a step in your process that could end up messing up everything and not getting you into the system that you're wanting to get into.

Below are some steps that you can begin with so that you can begin your journey to hacking.

1. First you are going to want start

using the Kali Linux system daily.

2. Next you will want learn algorithm and data structure on a deeper level. If you are in school, then this will most likely learn this in the second year of your computer classes. It is a good idea that you go a step further and learn both the data and algorithm using both the Python and C programming languages.
3. You are also going to need to have a very clear understanding of the operating system as well as the computer network. The most specific thing that you can have is a solid understanding of the memory management in the operating system

works as well as the process management along with cryptography, TCP/IP, and routing protocols within the computer network. It is best to know how to use these with Python as well as C. this is also a good time to learn Linux or UNIX.

4. There is a need for you to understand how websites work. In order for you to understand that, you are going to need to understand HTML, JavaScript and Apache. There also server languages that you'll need to understand such as PHP, CSS, Django, and My SQL along with several others.

5. Now that you have the understandings of how the computer networks and operating systems work, you are not ready to dip your toes into the hacking world a little further. Now you are going to want to have a clear understanding of the vulnerabilities as well as the attacks for programs such as SQL injection, LFS, RFS, XSS, Remote shell, Buffer overflow attacks, brute force attacks as well as being able to reverse the TCP payloads etc.
6. Now you're ready to be more hands on about testing your hacking skills. There are hacking tools that you can use to practice such as Wapiti,

sqlmap, Cain & Abel, Metasploit, airmon, and Aircracking. These programs are going to help you improve your hacking skills and will allow you to test different ways of hacking to find the method that works best for you.

7. Once you're comfortable with the skills that you have now worked hard to get, you're now ready to try and make your own hacking tools. This can be done by using programming languages such as C or even Python (Python is an easy to use programming language). The most important part of this step is for you to remember that you are not going to

create a tool such as Metasploit on your first try. You're going to have to take your time and keep practicing. Each time that you practice, you're going to get a little better. When you see that you have made a mistake, take a deep breath and just try it again.

8. Besides being patient, the next most important thing to remember is to not get in a hurry. Learning how to hack is not like learning how to read or ride a bike. Just because you can use someone else's tools does not necessarily mean that you are a hacker. You're simply using someone's tools to do the hacking.

You're going to need to have an excellent understanding of the operating systems, network systems, programming language so on and so forth in order to make yourself a great hacker. A hacker is nothing more than someone who has a solid foundation in computer science and is an excellent programmer.

9. It is also advised that you talk to someone who has been hacking for a while. Find someone who can walk you through the steps that they went through in order to become a hacker. If you're going to do this, you're going to want to find someone who is a senior at hacking so that you're

not going to someone who is just as new as you are. Go to someone who can help you understand the things that you do not understand and will help you in advancing your skills.

0. Last but not least, remember the quote from the Spiderman movies. “With great power comes great responsibility.” Be a good guy, not a bad guy. Do not destroy any resources that you have and do not use your newfound skills to harm others.

Chapter 2: Wireless Hacking Basics

When learning how to do something new, you always need to know the basics so that you can ensure that you have the knowledge that is necessary to do the job. Hacking is no different.

There are basics to hacking that you are going to want to cover before you delve into how to really hack a wireless system.

- An ad hoc network is going to be a network that will not allow you an access point for central coordination. All of the nodes that

are on an ad hoc network is connected peer to peer that way it is an independent service. They also have what is known as an SSID.

- Local areas that host wireless networks are based upon an IEEE 802.11. The IEEE 802.11 are the standards that are in places for wireless networks. These standards are written and enforced by the Institute of Electrical and Electronic Engineers.
- There are two kinds of networks, an infrastructure network and an ad hoc network. It is easier to use an infrastructure network because it has more access points allowing for

traffic to move seamlessly through the nodes.

- Every access point on a network has a basic service set which identifies the Mac address for that particular node.
- The extended service set is a character string known as ESSID.
- Basic sets work with one node on the client that's is using that network. The extended set works with several access points on the client at once.

Network frames

When using a network, you will be working off three different frames. These frames control the network and everything that is done on it. These

frames are the data frames, the control frames, and the management frames. Each frame is going to have its own function in making sure that the network is going to work properly.

- Data frames work with showing the real data that is on the network and you are going to be able to compare it to the frames that you find on Ethernet.
- Control frames make sure that what one client is doing is not messing with what another client is doing inside of the network's ether.
- Management frames ensure that the network is connected and is configured the way that it is

supposed to be. Not only that, but the management frames work with the reconnaissance that you are going to need to do on the network that you are wanting to hack into.

- The disassociation and deauthentication frames are going to tell the node that it has been authenticated or associated with the network and therefore a new node has to be made for the network to work properly.
- Beacon frames work best whenever you are trying to do reconnaissance on your target. The beacon frame is used to monitor how strong a signal is for a client from the point of

access that they are using.

- Association response frames allow clients to use the frame and see if they are able to get information on the network.
- Probe request frames are very similar to the beacon frames. There is going to be a request sent from the client to the node where it is wanting to connect to the network. There is going to be all the information needed that the client could want for the network that they are trying to connect to.

Chapter 3: Getting Information on the Target

Before you can even begin to think of hacking someone's network, you need to make sure that you are getting all of the information that you can on the target. Doing reconnaissance on your target is known as wardriving.

It is recommended that you use a laptop, the antenna that is on your car, power inverters, a wireless card, and a GPS receiver in order to connect to a wireless network. With all of this equipment you are going to be able to get any and all information that you need so

that you can get into the network without any issue.

Ensure that your laptop and wireless card are up to date enough that they can support rfmon or monitor mode.

Below are some programs that will assist you in getting the information that you need from your target.

Kismet

This network traffic analyzer is going to be best when you are using it on Linux, OS X, FreeBSD, or NetBSD systems. You can get this program for free and it has an open source. Kismet is one of the more popular programs that wardrivers use when they are serious about getting

into a wireless network because you are going to be able to see when the most traffic is going through on that network which is going to enable you to get on without anyone knowing that you are there. Or, so that you can get on when you do not have anyone else on the network taking up the memory space that you need.

Netstumbler

Netstumbler is a program that is for Windows and it is free. Again, this is a program that is popular for wardriving but it is also popular when someone is trying to get information on their target but there is a disadvantage to using this program. You are risking that you are

going to get caught when you use this program if the target is using a wireless intrusion detection system due to the fact that it is probing the network for the information that you are searching for.

Another thing that comes with Netstumbler is that it has a GPS unit that gets all the information and associates it with the proper networks that are discovered. In 2004 there was a new release for Netstumbler and it was discovered that it was not going to work with Windows XP or Windows Vista.

InSSIDer

Unlike Netstumbler, inSSIDer is going to work with Windows XP and Vista. It is also going to work with Windows 7, 8

and any Android products. This is another free program that has an open source. Like most of the other programs, it has a GPS device but it also has a wireless card or even a wireless USB that is going to run the program. The user interface for inSSIDer will show the SSID, the strength of the signal, the MAC address, what type of hardware is being used, the network type, and even the security that is on the network.

Wireshark

Wireshark tracks the traffic that is on a network and shows all of the packets that are on that network. It can be run on almost any operating system that you are going to be using. While you are going to

get a lot of useful information when it comes to using Wireshark, it is not going to be decoded or analyzed by the product, although you are going to be getting results that other products are not going to be able to get.

Androdumpper

This is an Android program that will test as well as aid in the hacking of a Wi-Fi router that is using WPS because all WPS have vulnerabilities and Androdumpper is going to hack the network with a series of algorithms.

AirMagnet

There are two different programs that you can get from AirMagnet; the laptop

analyzer and the handheld analyzer. Both of these programs are going to give you a full analysis of the network that you are targeting and the user interface is going to be simple for you to understand and use. But, it may not work well for someone who is trying to wardrive like some of the other programs that we have discussed.

Airopeek

With Airopeek you are going to locate the network packets and see the traffic that is occurring on that network.

Airopeek is going to work on almost any Windows product and is going to work with most of the network interface cards that you are going to be able to purchase.

In fact, Airopeek is used most often when trying to capture as well as analyze the traffic that is going through a wireless network. It also works a lot like Wireshark does.

Getting information on local networks in your area

Sniffing

With sniffing you are going to be able to locate different IP addresses which is going to assist you when it comes to mapping the network.

Footprinting

You are going to be enabled to find the reachable and relevant IP addresses to

what you are trying to get ahold of. This is usually what is used when you are trying to attack an organization over the internet. Relevant IP addresses are going to collect DNS host names and then translate them into an IP address and the range of that IP address, this process is called footprinting.

Search engines are going to be used so that you can find all the information that you need on your target. There are a lot of times that certain organizations are not going to have their resources protected from the internet because a web server has to be used so that they can use that tool. Then there are the various email servers and other parts of

the system that they are going to have to use that are going to make it to where they have to have access from the internet which is going to be the way that a hacker is going to get in. For organizations, the IP addresses are going to be grouped together, so all it takes is gaining access to one and then the rest are going to be able to be found.

A name server is going to have the domain names that either will be translated into the IP address or into the name server. When you are using a Windows system, you can use the NSLookup command to search for the DNS servers. As you enter words into the search, there is going to be a list of

commands that pops up so that you can tell the system what it is that you are wanting to do. Should you be using a Linux system, then the command is going to be used whenever you search for that DNS server before the list appears for all of the options that you are going to have access to. However, your -h option is going to be the only one you will want to invoke. With this command, you are going to be able to reverse the walker for DNS as well as the entries in the range that you are working with.

Host scanning

After you have successfully gotten into the wireless network, you are going to want to figure out what the topology is

for the network. This includes what the computers are going to be named and the number of computers that are on the network. The best program to use for this is Nmap which can be used on Linux as well as Windows. Sadly, it is not going to give you a network diagram.

When using a network scanner such as Network View, it is going to ask for the range of IP addresses or for one particular IP address, after you have inputted this information, you are going to allow the program to finish its scan before a map is displayed showing you all of the routers, laptops, servers, and work stations that are on the wireless network.

Chapter 4: Getting into a Wireless Network

There are several different ways that you can hack into a wireless network. In this chapter, you are going to learn what you need to do in order to get into the network and what methods can be used. The method that you use is going to be up to you based on your experience and how comfortable you feel with the programs that you are going to have to use in order to get into that network. For your own safety, for your wireless network that you use, you need to change the password so that it is harder to hack.

The password that comes with the router is too easy for someone to hack because it is the password that comes from the factory. However, when you change it to something that is personal to you, then you are going to be making it harder for someone to get into the network.

Aircrack-ng

You are going to find that Aircrack-ng is going to run best when you are using a Linux or Windows operating system. with Aircrack-ng you are going to be able to crack the WPA and the WEP that may be in place on that system. If you are launching a KoreK or Psychine-Tews-Weinmann attack Aircrack-ng is going to be one of the best options that

you can use. There are components from Airmon-ng that are located in the program that make it to where a wireless network card can be configured.

There are three different parts to Aircrack-ng that you are going to have to use in order to get the results that you desire. Airodump-ng which is going to give you information on all the frames that are being used in that particular network. Aireplay-ng which will show the traffic that is going through the network. And finally, Aircrack-ng is going to do the actual cracking of the network based off of the information that you got from the other parts of the program.

As for the packets being decrypted, airdecap-ng is going to take care of this.

CoWPAtty

You are going to start a dictionary attack against the network you are trying to hack. CoWPAtty works best on a Linux system. This program works off an interface that uses command lines so that it can find the word or phrase that will give you access to the network. Think of it as a handshake that is going to take place between four different components, the EAPOL as well as the SSID.

MAC address

You can use the MAC address as a way

to use a vulnerability to get into a wireless network. You can also use encryption if that is what you want to do. Ultimately, the MAC address is going to be changed so that it matches the client therefore the network is going to assume that you are that person and it will allow you into the network. When you work with MAC attacks you should be working off of a Windows system.

Void11

Working off of the Linux system, the Void11 program is going to deauthenticate the client to allow you into the wireless network.

Hacking Wi-Fi

Wireless networks are routers or any other way that a person or family gets Wi-Fi in their home or business. These can usually be easily hacked because a lot of people do not change the password to the router from the original password that is given by the wireless provider.

It is a good idea to always change the password that is provided on the router that transmits data into the location in which it is located. This will help to make it harder for hackers to get into your Wi-Fi therefore making it easier for them to get access to yourself.

Step one: make sure that you have the appropriate programs downloaded.

There are going to be two different programs you'll need in order to make this hack work. CommView and AirCrackNG will help you to look for vulnerabilities in the network as well as help you to break the security key.

Note: make sure that your computer's wireless adaptor is actually compatible with CommView

Step two: now you need to find a network. CommView is actually going to scan for any wireless networks that it can find. All you need to do is to select a network that has a WEP key and a decent signal.

Step three: filter your search to that network specifically. Right click on

which network you are wanting to use and select the copy MAC address. From there you'll go to the rules tab and down to MAC Addresses. You'll enable the MAC address rule, click action, capture, add record, both before you paste in the MAC address.

Step four: from here you'll need to sort out the management and control packets so that you are only viewing the data packets.

Step five: by going to the logging tab, you are able to enable the auto saving mode. You may need to go and change the settings on the directory size and file size. You can try 2000 and 20.

Step six: now press the play button so

that you can begin collecting. You're going to have to wait until you have about 1000,000 packets.

Step seven: at this point in time you need to click concatenate logs to make sure that all the logs are selected.

Step eight: export the logs. You're going to go to the folder where your logs are saved and open it. Next you'll click on file and export then select WireShark/tcpdump format and save it so you can find it easily at a later date.

Step nine: open the newly created file with Air Crack.

Step ten: enter your index number. When your command prompt opens, you're going to need to enter the index

number for the network that you are trying to target. It is most likely going to be one. Once you've done this, you'll hit enter and wait. If it works, then the key will be shown on your screen

Hacking scenarios

Scenario 1: There is a computer that has no encryption on it which means that the network is wide open. Therefore, there is no isolation for the client and the network is considered to be unsafe to use and easy to hack.

Scenario 2: WEP (the key that is provided by the router's provider) is being used. There are several known attacks that exist and it will then make it easy to hack the network.

Scenario 3: The computer is not encrypted except for the isolation is enabled and a captive portal exists. With this type of wireless network, it is acceptable for a visitor to use the internet. Therefore, it should not be used for a company as it is still easy to be hacked.

Scenario 4: WPA (Wi-Fi protected access)/WPA2 is being used and a strong password has been put to use. The password has sixty characters, lower-case, upper-case, no dictionary words, and special characters in it. A hacker would not be able to crack the password with any computing power that we currently have. However, if the

password is not changed every three months, there is a likelihood that a hacker will be able to figure out the password.

Scenario 5: WPA/WPA2, a weak password has been chosen. A hacker can now capture the authentication handshake and then make some attempts to crack it by using his own machine or even a “cloud” server. The server can then be compromised within a minute all the way up to a few hours.

Scenario 6: A company is using a WPA and a strong password that they change every day. But, the router that they are using in order to transmit WiFi has a static WPS pin that they are not able to

change or even disable.

Because WPS is enabled, this is very similar to having an open network. So, this network is considered to be unsafe and should not be used for business purposes.

Scenario 7: RADIUS is being used and the settings are weak when it comes to the wireless clients and the server. A hacker would be able to perform what is called a rouge AP attack and obtain the authentication handshake.

Should a weak password also be used, it can be captured and user accounts will be at risk as well as the network being compromised. It is important for each person on this type of network to have

their own password that is tied directly to the domain. This means that the hacker will not be able to hack the wireless network as well as the domain.

Scenario 8: The company is using WPA/WPA2, as well as a strong password that is changed every day. The WPS is disabled and the administrator's computer is kept up to date.

But, the router has not been updated since being installed and it contains odays (unknown vulnerabilities) that will allow a hacker to be able to conduct a CSRF attack. This is done by a persistent threat and the following can happen:

- The router will be compromised

- The hacker will be able to send targeted emails within the system administrators system that will cause it to appear like it is being sent from the router vendor. This will also inform the system administrator to log into the router and check the email by clicking a link within the email after they have logged in.
- The link will then redirect the administrator to a page that will change the routers settings or simply steal the password.

It is also possible for a hacker to be able to get into a system because an employee has shared the password to the system unknowingly with a hacker and then makes the system compromised. This

can also happen knowingly.

Or, if an employee's phone or computer is compromised, then the wireless network password is compromised as well.

You should have a strict ACLs from the wireless to any segment that is wired. There should also be strict ACLs to any server that is going to hold sensitive information.

Chapter 5: Scanning Ports

The whole reason you are going to want to scan ports is so that you can find an open one. With ports, you are going to be able to get into someone's system and leave a door open so that you are able to get in again later on. Port scans use host scans which can take up a lot of time if you have a wide range of IP addresses that have to be scanned and most of them end up being vacant.

Ports that are open

With some of the programs that we mentioned above, you are going to be able to use your internet connection to

use a protocol of either TCP or UDP. These protocols are going to help you see what ports are on the system that you are trying to gain access to. Ports allow for programs to run all inside of a single IP address. You will discover that most programs work off of a default port. For example, an HTTP server is going to use port 80 with the TCP protocol.

A network scanner is going to be used when you are connected to either of these ports and as soon as the port accepts the connection from the scanner, it is going to be best for you to assume that the program that is bound is running as it should be.

TCP ports are going to work with SYN

packets that are sent back and forth between the servers and the clients use them. Whenever the packet is sent to the server, it is going to send a SYN/ACK packet back resulting in the client sending the ACK packet back. After the SYN packet is received once more by the client, the port is going to be opened. In the off chance that an RST packet is sent instead, then the port is going to be closed. If the server does not send anything then there is probably a firewall that is blocking it from the port or the port is not running on that IP address.

When you are scanning UDP ports, you are going to most likely run into

problems because there are no handshakes exchanged and the programs are going to get rid of any packets that they are not going to be able to process. UDP packets are going to be sent to a port without a program that is bound to it. ICMP error packets are going to be what is returned. From there you are most likely going to consider the port to be closed. No answer is going to mean that a firewall is filtering out the packets or the port is opened. Too many people end up leaving their UDP scans because these scanners have difficulty telling the difference between when a port is opened and when it is filtering the packets.

Ports that are more common

In order to save yourself some time, Nmap is going to scan around 1667 ports that are going to be the default ports.

But, you are going to get more results if you thoroughly scan all the ports; and there are 65536 ports. So, if you have the time, scan them all!

Port specifications

When you are using the -p command, you are going to be able to tell the Nmap program exactly which ports you want it to scan so that you can save time on your scanning.

Target specifications

Just like you can tell Nmap to scan

specific ports, you can also tell it to go after a specific host or set of hosts. This host is going to be verified only by putting in the IP address for that host or by using the domain name. Should you wish to scan several different ports, you are going to want to set up the range for the IP addresses.

Scan types

TCP SYN

A TCP SYN scan is going to be the default scan done by Nmap. When you use the `-sS` command, the program will only do that scan. As the administrator, you are going to be allowed to start the scan. If a user starts the scan, then a

connect scan is going to be performed.

TCP connect

There is a command that you can use in order to make sure that Nmap has full connection and that is the -sT command. This scan is not going to be as good as the TCP SYN scan because there is more that has to be sent back and forth between the client and the server. This scan is going to be executed with user privileges or whenever an IPv6 address is being scanned.

TCP null

When you use the -sN option, the program is going to send back all packets that do not have anything to do

with SYN, ACK, or RST flags. If it comes back that the port is closed, the RST packet is going to be the one returned. If the port is opened or has a firewall filtering its packets, then there is not going to be a response sent back. Doing a null scan is going to be the best way to attempt to get passed the stateless firewall however if the firewall is stateful then it is not going to do anything.

UDP empty packet

When you use the -sU function, Nmap is going to send out UDP packets that contain no data. If an error message is returned, then you are going to assume that the port is closed. However, when

there is no response, you will assume the port is opened or filtered. However, this scan cannot tell the difference between a filtered port or an open port which is going to leave some severe limitation in your scan.

UDP application

You are going to use -sU or -sV options to tell the program that you are wanting data from an application or for the application to be identified. Since this is several different options put together, you are going to experience a slow scan.

Scanning speed

Like most things, if things are sent at a speed that is faster than the system is

able to deal with, then the packets are going to be dropped and they are not going to be used in the scan thus you are going to get results that are not accurate. If there is an intrusion detection or an intrusion prevention that is in place on the target's network, then the faster that the scan is going through the more likely that it is that you are going to be detected by the target.

There are a lot of devices as well as firewalls that work with IPS that are meant to respond to SYN packets that are sent in from the cookies created by these packets so that every port appears open even if they are not. When you are running a scan at full speed, then you are

going to risk wreaking havoc on the network devices that are stateful.

With Nmap there will be five templates that you can use in order to adjust the speed in case it does not adjust itself properly. With the -T0 option, you are going to force the program to wait about five minutes in between sending packets. -T1 waits for fifteen seconds, -T2 for 0.4 seconds, and -T3 which is going to be the default setting where the timing goes unchanged. Lastly, when -T4 is used, the time outs are reduced but the retransmission speed is upped ever so slightly. -T5 is similar to -T4 but things are going to be sped up even more. A modern IPS or IDS device is going to

figure out the scans that are using -T1 and detect that device so that the hacker is discovered. As the user of Nmap, you can also decide to make a new template with new parameters if you are not happy with the ones that are provided.

Identifying applications

If you decide to use the -sV option, then Nmap is going to have to figure out which version of the application is currently being run.

Identifying the operating system

If you want to discover which operating system is being used by the target, you will use the -O option in Nmap. There

are packets that are specially crafted to be sent to the target to all of the ports so that the responses can be analyzed in the database that you are using on your own operating system.

Save

When you want to save the output that you get returned to you, you will use the `-oX<filename>` option so that it is saved in an XML format.

Chapter 6: Vulnerabilities

It does not matter how secure a network is supposed to be, there are going to be vulnerabilities that you can use to get into the wireless network. Most of the time, a vulnerability is going to be a bug that is inside of an application that is affecting the security that you have in place to protect yourself. You can find these bugs on applications such as BugTraq. The CERT (Computer Emergency Response Team) puts out a report every year that tells you how many vulnerabilities they find so that people can better protect themselves.

Vulnerability scanning

When you can for vulnerabilities, you are going to be looking for any known vulnerabilities that you may be able to exploit on your targets network.

Nikto

With Nikto you are going to be scanning the web so that you can find applications that have weak spots along with files that might be dangerous. With this open sourced software, you are going to be able to find a version that works with either a Linux system or a Windows system. when you are using this program, you will be using an interface that works off of command lines.

Nessus

You have probably heard of Nessus since it is one of the vulnerability scanners that is known around the world. You are going to be able to use Nessus for free and it can work on almost any operating system. there are plug ins that Nessus uses that are going to assist in finding the vulnerabilities depending on the sort of bug that you are wanting. However, you need to make sure that you keep your plug ins updated.

There are also non-intrusive scans that you can do with Nessus that is not going to harm the target like an intrusive scan would. These scans are going to require that you have the domain name or at least

the IP address for your target. With this program, you are going to be able to scan the ports so that you can determine which programs are running on that network as well as the operating systems that are being used.

After the scan, has been finished, a report is going to show all the ports that were found to be open and what their vulnerabilities are.

Exploiting vulnerabilities

When you take advantage of a bug that is inside of an application, then you are going to be sending various commands out that are going to be executed to prevent the program from running the way that it is supposed to run. You can

do thinks like pass by the authentication that you may need to get onto the network, get more privileges than what you currently have access to and more.

Metasploit

This framework was first released in 2003 and had a specific set of things that it allowed the user to do to their target.

These things were:

- Integrating the evasion and encoding process.
- Making sure that a single database could be exploited through the use of easy updating.
- Having an interface that had options
- And combining the exploits with payloads.

All of these things take place whenever:

- You use evasion to bypass the security on a device through employing evasion techniques.
- There is a code that is used to exploit the module where the code is located so that specific vulnerability can be used.
- You have to modify the encoding for the payload you receive so that you can avoid the limitations that are cause because of the vulnerability that was located.
- Your payload has a code that has to be sent to a different location so that the action can be taken on the vulnerability.

- When you need to use specific options so that you can select what is hit by the payloads and the exploits.

Using Metasploit is pretty simple because you are going to be following the same basic set of procedures each time you use it.

1. Decide which exploit you want to use
2. Set up your payload
3. Choose the IP address you are targeting as well as which port you are going to gain entry through
4. Execute your plan
5. Evaluate your results
6. Decide if you can start or restart

your procedure

If you are trying to find the vulnerabilities that are inside of a host, then you are not going to want to use Metasploit instead, you will want to use a scanner that is mean to find all vulnerabilities in the network. If you do not want to do that, then you can always use a port scanner so that you can find the open points and exploit that.

With version 3.0 you will have a few different payloads that you can run with when you are working with vulnerabilities.

- Meterpreter: with this payload you are going to be using a command line interface that is going to run

specifically on Windows.

- VNC injection: This also runs on Windows, but you are going to get a graphical interface to your target so that is going to be synchronized with the user interface that your target is using.
- Add user: when you add a user, you are going to need to have a specific name and password and the account is going to be required to have administrator permission.
- File execution: a file is going to be uploaded on the targets computer and then the file will be run thus running any malicious code that might be inside of the file.

- Interactive shell: there is going to be another command interface that interacts with the target carrying out any commands that you give it.

When working with a VNC connection, you should ensure that you have a large enough bandwidth so that your program is running the way that it is supposed to. Along with that, you do not want someone to be in front of the computer that you are trying to hack. On the off chance that someone is there, then they just have to interact with the program you have running and notice that you are doing something to their computer.

OS X and Linux are going to be using the command line interfaces that are more

powerful than the ones that are running off of Windows. Just like anything else, the program also has its disadvantages.

Keeping control

The whole point behind hacking into someone's network, is to get control of their system. But, the best thing that you can do is to keep the privileges that you gave yourself for their network. Once you have made your way into the program, you are going to want to install a rootkit onto that computer so that you can have maximum control over the network.

Be careful though because there are a few programs that you may use that are going to end up compromising the new

accounts or computers that are found to be listed on the network. However, there are a few programs that are going to hide the fact that you are even there. When you are using these kinds of programs, they may make it to where there is a false version of the network that you have hacked using tools like netstat. Even further, there are programs that are going to remove any data that you may leave behind on the computer so that you can ensure that you are not going to get caught.

Depending on which rootkit you are using is going to depend on if you get any passwords that may be travelling over the network. You may also find that

you are going to have the ability to get in and modify the operating system that the target is using. If you do have this ability, you need to make sure that you are being careful because you do not want to let your target know that you are on or have been on their computer.

Back doors

As you get into a network, you may want to create a back door so that you do not have to work so hard when you are locating the system administrators because they are going to make it to where you cannot log or monitor the results that are going to come out of a normal network. When you are using a back door, you will be able to conceal

the accounts and which privileges that you have so that the target cannot see how far you have gotten. There are programs like Telnet that is going to make it so that you cannot have remote access in order to configure and operate as you wish.

The biggest reason that you are going to want to use a back door is so that you can keep the communication open between the target and your computer. Many of the methods that you are going to use are going to be things such as transferring files and then executing the program that is inside of the file. Make sure that any communication that you have with the target's computer stays

secret and make your back door secret so that other hackers are not using your entry point to the network.

A program called Back Orifice 2000 was made specifically to be a back door on a network. The sever for this program will run on Windows but the clients for it are going to run on Windows, Linux, and most other operating systems. Your server is going to be able to be configured so that you can use it as a utility. Once you have configured the server, then you should upload it to the target before you get started. Back Orifice 2000 makes it to where you can execute files, log keystrokes, transfer files, and even have control of the

networks that are on the network.

The AES plug in is used when you are dealing with traffic that is encrypted while the STCPIO plug in is going to be for the obfuscation of the traffic that is occurring on that network.

Rootkits

Rootkits are best for hiding your activity and other programs that you are using on someone's network.

The Hacker Defender is a rootkit that is going to be used on Windows. You are going to be hiding files and all of the things that come with it so that the target cannot figure out that you are there. You can use rootkits as a back door with the command line interface however the best

thing that you can use it for is to hide your files on your targets computer.

Chapter 7: Protecting Yourself and Preventing a Hacker from Getting In

As you have noticed throughout the content in this book, when a network is unprotected, it is going to be insecure therefore anyone can get onto the network and get your information or whatever else that they may be wanting to get their hands on. However, the IEEE had made a standard for making sure that wireless networks come with a WEP. With WEP you are going to have a security protocol that is put into place which will:

- Integrity: the data cannot be altered and is going to be exactly as it was when the target left their network.
- Authentication: all the users that are on the network are going to be able to be identified to make sure that they are allowed to use the network.
- Confidentiality: anyone that may be trying to get onto the network and get gather information such as passwords are going to be protected.

There are a lot of experts in security that criticized WEP while a great majority of them now find it as ineffective.

When 2004 came around, a new protocol was drafted and sometime around 2007, and it included the newest

standard by IEEE. This is where WPA2 came into play. WPA2 has a block cipher rather than an algorithm so that it can be used for key distribution along with authentication. Thus, making WPA2 more secure, despite that, WEP is still being used.

Most wireless routers that people use are going to control the MAC address so that it can authorize that the wireless network is authentic. MAC addresses are going to be what keeps the entire neighborhood from using your network slowing your connection speed down and making it to where they may end up getting ahold of information that they do not need to have. Sadly though, it is not

going to stop a hacker that has the experience needed to get past this security measure, let alone one that can use MAC addresses to get into the network.

Whenever you try and turn off broadcasting for your SSID, you were adding some extra security to your network, however, this was far from true. a simple search on the internet will help a hacker locate programs that are going to reveal the SSID on your computer even if it is not broadcasting. The Microsoft company eventually came to the conclusion that when the broadcasting is turned off for a computers SSID leaves less security on

a network making it easier for hackers to get into a network.

Looking at encryption, the specifications for WEP no matter what the strength of the encryption is going to make it to where it can withstand hacking, even from the most determined hacker. This is way WPA was first created as a way to try and allow for more protection on networks. There are upgrades that become available often when one is using WPA unlike with WEP. With technology constantly evolving, security had to evolve as well and thus, WPA2 was born so that it can be supported on the newer technology. Anyone who works with security is going to

recommend that you use hardware that is only going to support the WPA and WPA2 security protocols.

Also, ensure that you are installing any updates that may come across your computer. You should also create a strong password, customize your SSID, and disable the WPS. Should there flaws in the security that go unpatched, then a hacker is still going to have access to get into your network and get any kind of information that they desire. In the event that you notice your router provider is not giving you updates in a timely manner or not at all, then you need to look into either changing models of router or finding a new provider.

Detecting a security breach

As we have discussed in previous chapters, there are things such as sniffers or network scanners that are going to be used with a network interface card. The card is going to be tuned with a set number of radio channels. When a passive scanner is being used, the scanning will not be detected by the target.

Being a hacker, you are going to be able to get massive amounts of information from your target all through the use of a passive scanner. However, even more information can be obtained when crafted frames are used to get more useful results from your target. When you

send out crafted frames you are going to be doing what is known as active scanning through the use of a transmitter that is inside of your wireless card. You need to be careful with this method though because you are going to be able to be located all because of your wireless card.

Being detected is not going to be something that you are going to want because you are going to end up getting in trouble. A target that believes that they have been hacked can get the proper programs and equipment that will begin to track your moves and everything that you have done on their wireless network all while transmitting back where you

are located so that your target knows where you are and can turn you into the police.

Chapter 8: Hacking Techniques

It really does not matter what you are trying to hack into, there are going to be techniques that you are going to follow in order to make sure that your hack is successful.

1. Anonymity: Hackers don't want you knowing that they got into your system. In doing this, they are going to make sure that they do anything that they can in order to not leave a trace. In doing this, they will use
 - Proxies or secured tunnels
 - Software that will hide their IP

address

- Will use other people ' s usernames and passwords without using his own
- Programming that is written in C
- Or a telnet that will hide him and his task execution.

2. Getting Out: Any good hacker is going to make sure that they do not leave any trace on your computer so that you never know they were there. In the process of getting out the hacker is going to leave all your files alone, however, he is going to leave a “ backdoor ” open so that he is able to get back into your computer at a later date.

3. Gather information about the target:
When hacking someone, it is important to get as much information on them as you possibly can. In doing this, you ' re going to want information such as:
 - Their IP address
 - Telnet or Tracert in order to look at the pings of when someone is on the computer
 - And be resourceful. Find out as much about your target as you possibly can get.
4. Log the keystrokes: as a hacker, you are able to use programs that will review every keystroke that someone has made on their computer and that

alone can reveal a person's identity.

5. Go for passwords: when trying to hack passwords, it is best to try and go with the simple algorithms that create a combination of letters, symbols, and numbers. This is a trial and error method. You'll need to make well educated guesses and use dictionary attacks so that you can generate every possible combination of the password.
6. Leave a virus: this is a simple way to get back into a computer. Leaving a virus can be done by simply sending out an email or instant message to any potential victims.
7. Gain entry through a backdoor: this is extremely similar to hacking a

password. There are many hackers who develop codes and programs that will look for defenseless pathways into networks as a way to enter the network without ever having to use a password.

8. Spy on email: this is a program that can be used so that you can interrupt and read emails.
9. Make zombie computers: this is when a computer is used by a hacker to place DDoS attacks and send out spam emails. If an innocent user clicks on the link that you send out, it will open up a connection between your computer and his.
0. Make sure that you have a firewall up so that you have restricted access

to any personal information that may go outside of your computer.

1. Use a proxy server: when you decide to use the computer, make sure that you target proxy servers.
2. Use search engines: this will help you to find the tools that you need in order to hack a system. From here, you can download the tools that you need in order to target a specific computer.
3. Leave a file or two on the computer: this will allow you to gain easy access back into the computer at a later date. You can leave files such as Net cat in order to gain access again.

Chapter 9: Types of Hackers

Just like with anything else, there are different levels of hacking. These hacking differences cause each different hacker to fall into a hacking category. Each hacker is on a different level of their actual category but each category falls into the good, bad, and somewhere in between categories.

The three types of hackers fall into the white hat hackers, gray hat hackers, and even black hat hackers.

In this chapter, we're going to go over what makes each category of hacker into a hacker.

White Hat Hacker

This is a term that actually refers to a hacker who is an ethical computer hacker or even a computer security expert. These hackers usually specialize in penetration testing as well as other methods that they can use in order to ensure the security of an organizations security system.

Ethical hacking was actually a term that was coined by the IBM in which it simply meant that it was to be implied to a broader category than just those who used penetration testing.

As opposed to those who are considered to be a black hat hacker, a white hat hacker is someone who does not use

hacking for malicious intent. The names were actually coined after the old Western films where the heroes wore white hats and the antagonistic wore black hats in order to be distinguished between good and bad.

White hat hackers are also known as sneakers, tiger teams, or even red teams. Those who also do phreaking are known as white hat hackers.

Some of the most well known white hat hackers are: Kevin Poulsen, Kevin Mitnick, Robert Tappan Morris, Barnaby Jack, Michael Mansfield, Raphael Gray, Eric Corley, and Przemyslaw Frasnuk.

Black Hat Hackers

These hackers are normally known to those who violate a computer's security system for little reason more than to be malicious or to gain access to personal information for personal gain.

Richard Staliman is the first person who actually used the term black hat when speaking about a criminal hacker versus those who just hack for their own general curiosity.

Black hats are what society generally thinks of when they think about hackers. These are the hackers who are portrayed as those that are the "epitome of all that the public fears in a computer criminal." These types of hackers will generally break into any secure network in order

to modify, steal, or destroy data. They even make the network unstable so those who are authorized to use it are at risk.

Those who use cryptovirology are those who are known as black hat hackers.

Gray Hat Hackers

Gray hats typically fall between the black hat hackers and the white hat hackers. They typically are those who can hack a computer system that sometimes happens to violate the laws and ethical standards but does not do it to be malicious like a black hat hacker.

This term was first brought around during the 1990s in order to place a new name for these hackers so that they were

not considered to be white hats or black hats. Another way to tell the difference between these two hackers is to know see the different methods that they use in order to discover the vulnerabilities within a computer system.

Typically, a gray hat hacker will charge a small fee to the company in which they find a small vulnerability in their system in order to fix the hole in the system or network. When you look at the SEO (search engine optimization) community, a gray hat hacker is defined as someone who usually manipulates a web site's rankings on a search engine by using improper or even unethical means that are not considered search engine spam.

Chapter 10: Hacking- The Effects Everyone Suffers From

When you hack into a computer, you are opening up the system that you are hacking to multiple effects. As the public sees, many of the effects of hacking are going to be bad because the public doesn't always see that hacking is something that is a good thing.

Just like with anything, there are pros and cons to everything and hacking is truly no different. When you hack someone, you are always leaving them open to the malicious effects of what

hacking brings about whether you mean to or not.

When you do hack a computer, you are creating a breach in the computer's security. This is placing the victim's sensitive data and privacy at risk. These hacking activities are usually done in order to gain access to confidential information that one tends to keep on their network such as: social security numbers, bank account data, credit card numbers, and personal photographs.

There are a few hackers that tend to use this information to harm the one that they have hacked, then there are others who simply "take" this information in order to prove to their "victim" the major

security issues that they have in order to get them to be fixed so that the personal and sensitive information that they have taken cannot be taken again.

Once a computer's security system has been compromised, there is also the possibility of the loss or even manipulation of data. A hacker can go in and delete any sensitive information that has been placed on a network once they have gained access to it. Once your system has been hacked, you're at a great risk for all the data that you have on your computer being lost or manipulated in a way that can and most likely will harm you.

One of the biggest things that everyone

associates with hackers is identity theft. Identity theft is when someone who is not authorized takes your identity. Your identity is but not conclusive to your social security number, date of birth, or anything else that would identify you as you. This is usually done with a malicious intent and used for the hacker's personal gain or interest.

When you've been hacked, the hacker can actually track everything that you do on your computer thanks to the advances in technology. Key-logging software is what is used in order to track every keystroke that you make on your computer. Thanks to this software, a hacker can instantly gain access to your

passwords, your bank accounts, and anything else that they can use to harm you, all thanks to one little program.

DOS means denial of service attack. This happens when a hacker gets into your network and your computer therefore makes computer resources unavailable to any of the authorized users. It is most often that a DOS will attack a website which will then make the website unavailable for a long period of time. This then causes the users of the website to be inconvenienced as well as hampering with the business of the website.

Along with identity theft, stolen information is a big thing that happens

when someone hacks your network for a malicious reason. This can be hazardous to anyone, but most particularly to business' that end up being hacked because then sensitive information that they do not want getting out to the public is then released. Not only that, but email address', client information, so on and so forth can be stolen and compromised. National security can even be put at risk when it comes to hacking. Hackers who hack into the governments networks then have access to the defense system as well as many other systems that will cause there to be grave consequences on the welfare of the nation. When someone hacks into the government, not only is the

nation's security at risk, but so is the well-being of the citizens of the United States.

Another effect of hacking is fraud. Hackers can turn computers into zombies by infecting them with internet enabled computer viruses. These computers are then used for activities that are considered fraud such as spamming and phishing attacks on other networks.

How do you know when you have been hacked? Your computer will most likely decrease in its performance speed. You may also begin to notice files that are not supposed to be there. These files may increase in size as well as be modified without you ever touching them. You may

also begin to notice that there are changes in your network settings or even frequent disk crashes.

The only way that you're going to be able to protect your computer is by installing a reliable antivirus software as well as making sure that your firewall is enabled before you begin to connect to the internet. Also make sure that you install the system updates on a regular basis.